

AT29LV512@TSOP32.BIN - Firmware Analysis Report

Converted from the firmware analysis notes generated for the AT29LV512 TSOP32 ROM dump.

Item	Value
Image size	65,536 bytes / 0x10000
Architecture	Motorola 68000-style firmware
Reset vector	SP 0x00FF95FE, PC 0x0000042C
Identifiers	SunDisk, SDP3B, Rev 1.21
Likely device	SanDisk/SunDisk SDP3B PCMCIA ATA FlashDisk controller

File identity

- Size: 65,536 bytes / 0x10000, matching a 64 KiB AT29LV512 flash dump.
- MD5: 5b71767d855e07ee49049b5afd27b7cd
- SHA1: ab42eb6aa914cb70dbfa378f5ef5d46fdbad6557
- SHA256: 5afc4d8a361404bd7da0cf13b64e343876991442ac045a309f76781bbfc84610
- CRC32: db8ac5b6
- Last non-0xFF byte: 0xAAEB. Bytes 0xAAEC-0xFFFF are all 0xFF, so about 21,780 bytes of the 64 KiB chip are unused/fill.
- Overall entropy: 5.076 bits/byte; used region entropy: 6.308 bits/byte. No ZIP/gzip/PNG/ELF/PE/PDF/JPEG/etc. signatures found.

Architecture / boot shape

The image begins with a Motorola 68000-style vector table:

- Initial supervisor stack pointer: 0x00FF95FE
- Reset PC: 0x0000042C
- Main exception vectors point into 0x6C02-0x6E9A.
- IRQ levels 1, 2, and 3 all point to 0x00006978.
- Most user vectors 64-255 point to 0x00006E9A.

Reset routine at 0x042C, key instructions:

```
00042c: ori.w    #$2700, sr          ; supervisor mode, interrupts masked
000430: move.b   #$1, $8056.w
000436: movea.l  #$ff95fe, a7        ; supervisor stack
00043c: movea.l  #$ff95fc, a6
000442: move     a6, usp             ; user stack
00044a: movea.w  #$9000, a1
00044e: movea.w  #$9b00, a2
000452: movea.l  #$0, a0
000458-000462: loop clearing 0x9000-0x9AFF RAM/global area
000464: jsr      $5952.l
00046a: jsr      $6430.l             ; returns error if pointer check fails
000474: jsr      $59BE.l
00047a: jsr      $5B54.l
000480: jsr      $5AFC.l
000486: jsr      $5C10.l
00048c: jsr      $5A3E.l
000492: jmp      $0542.l             ; enters main loop/dispatcher
```

Exception/interrupt handling

The handlers at 0x6C02 onward all save D0-D2/A0-A1, put the vector number in D6, and jump to 0x6C00. Address 0x6C00 is an infinite branch-to-self, so unhandled exceptions deliberately halt the controller while leaving the vector number in D6 for debugging/postmortem.

IRQ levels 1-3 use 0x6978. That routine manipulates many 0x80xx registers, especially 0x800B, 0x8059, 0x805B, 0x805D, and 0x8056. It looks like the external host/PCMCIA/IDE interface service routine.

Strings / embedded identifiers

Meaningful strings found:

- 0x5246: SunDisk
- 0x524E: SDP3B
- 0x53D4: Rev 1.21SunDisk SDP3B-0
- 0x566A: COPYRIGHT (C) 1993,1994,1995 SUNDISK CORPORATION

The "SunDisk" spelling is historically consistent with early SanDisk branding. The ROM identifies itself as firmware for an SDP3B card family device.

PCMCIA CIS-like tuple block

A meaningful Card Information Structure tuple stream starts at 0x5242:

- 0x5242: CISTPL_VERS_1, len 0x15: major=0, minor=2, strings: "SunDisk", "SDP3B", "0.1"
- 0x5259: CISTPL_JEDEC_C, len 0x02: DF 01
- 0x525D: CISTPL_CONFIG, len 0x05: 01 0B 00 02 0F
- 0x5264..0x52B9: multiple CISTPL_CFTABLE_ENTRY tuples
- 0x52BA: CISTPL_DEVICE_OC
- 0x52C0: CISTPL_MANFID, len 0x04: 45 00 01 01 => manufacturer 0x0045, product 0x0101
- 0x52C6: CISTPL_FUNCID, len 0x02: 04 01
- 0x52CA/0x52CE: CISTPL_FUNCE tuples
- 0x52D9: CISTPL_END

ATA Identify Drive template

There is a 512-byte Identify Drive template starting at 0x53A6. The firmware copies it to the RAM buffer at 0x9600 during Identify Drive handling and patches runtime fields.

Notable fields using normal byte order in the ROM image:

- Words 23-26: firmware revision = "Rev 1.21"
- Words 27-46: model number = "SunDisk SDP3B-0"
- Word 49 template value includes 0x0200, which matches LBA-capable style capability bits.

Command dispatch observations

The main dispatch path starts around 0x0764. It reads the host command register at 0x8044, dispatches first on the high nibble using a PC-relative jump table at 0x07BE, then for several command families indexes 16-entry handler tables at 0x4F3C, 0x4FBC, 0x503C, 0x50BC, 0x513C, and 0x51BC.

Recognizable exact command checks / table entries include:

- 0x03 Request Sense: handler 0x0DD8 checks `cmpi.b #\$03, \$8044.w`.
- 0x20/0x21 Read Sector(s): table 0x4F3C entries 0/1 -> handler 0x0D52.
- 0x22/0x23 Read Long Sector: table 0x4F3C entries 2/3 -> handler 0x0D52 with different flag word.
- 0x30/0x31 Write Sector(s): table 0x4FBC entries 0/1 -> handler 0x101A.
- 0x32/0x33 Write Long Sector: table 0x4FBC entries 2/3 -> handler 0x101A with different flag word.
- 0x38 Write Sector(s) without Erase: table 0x4FBC entry 8 -> handler 0x101A.
- 0x3C Write Verify: table 0x4FBC entry C -> handler 0x101A.
- 0x40/0x41 Read Verify: special high-nibble 4 path, only accepts commands where command & 0x0E == 0.
- 0x50 Format Track: exact check at 0x0832, handler 0x0A68.
- 0x7X Seek: high-nibble 7 path, handler 0x1160.
- 0x87 Translate Sector: exact check at 0x0866, then dispatches through table 0x503C.
- 0x90 Execute Drive Diagnostic: table 0x50BC entry 0 -> handler 0x09C0.
- 0x91 Initialize Drive Parameters: table 0x50BC entry 1 -> handler 0x0D14.
- 0xC0 Erase Sector(s): table 0x513C entry 0 -> handler 0x0A50.
- 0xC4 Read Multiple: table 0x513C entry 4 -> handler 0x0DC2.
- 0xC5 Write Multiple: table 0x513C entry 5 -> handler 0x108E.
- 0xC6 Set Multiple Mode: table 0x513C entry 6 -> handler 0x0FFA.
- 0xCD Write Multiple without Erase: table 0x513C entry D -> handler 0x108E.
- 0xEC Identify Drive: table 0x51BC entry C -> handler 0x0AB4. It copies 0x53A6 to 0x9600 and updates capacity fields.
- 0xEF Set Features: table 0x51BC entry F -> handler 0x0E50.
- 0xF5 Wear Level: high-nibble F path calls 0x0E38, which only accepts command 0xF5.

Memory map inferred from code references

- 0x0000-0x03FF: 68k vector table.
- 0x042C onward: reset/startup and main dispatcher code.
- 0x4F3C-0x51FF: ATA command handler tables.
- 0x5242-0x52D9: PCMCIA CIS tuples.
- 0x53A6-0x55A5: ATA Identify Drive template / mostly zero-padded after live fields.
- 0x566A: copyright string.
- 0x8000-0x805D: heavily referenced hardware/ASIC/ATA/PCMCIA registers.
- 0x9000-0x9AFF: RAM/global state cleared at reset.
- 0x9600: sector/data buffer used for Identify Drive and transfer operations.
- 0xAAEC-0xFFFF: unused 0xFF fill.

What it probably is

This dump is very likely controller firmware for a SanDisk/SunDisk SDP3B PCMCIA ATA FlashDisk. It implements the ATA-style command set, exposes PCMCIA CIS information, maintains an Identify Drive response, services host-interface interrupts, and manages lower-level flash media operations through an on-card controller/ASIC.

Limitations

This is not a full decompile. The attached disassembly file is a Capstone-assisted 68k reachable-code disassembly seeded from the vector table and reset path. Some code/data boundaries may still need manual cleanup in Ghidra/IDA/radare2 configured as M68000 big-endian raw binary loaded at base 0x00000000.